

Dr. Anh Le-Duy Pham

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Curriculum Vitae • Updated September 2026

Research Profile

Chemical oceanographer and marine biogeochemist specializing in **ocean trace-metal cycling, ocean biogeochemical processes and marine ecosystems, and biogeochemical modeling**. My research combines observations, numerical models, and data synthesis to investigate how redox reactions, organic ligands, sedimentary inputs, particle dynamics, and ocean circulation regulate the distribution and biological availability of iron and other nutrients. I develop and apply regional and global ocean biogeochemical models to connect chemical processes with marine ecosystems, carbon cycling, and climate change.

Education

2014–2019 Ph.D. in Oceanography, School of Earth & Atmospheric Sciences, Georgia Institute of Technology

2009–2013 B.S. in Marine Engineering, Department of Meteorology–Hydrology–Oceanography, Hanoi University of Science

Research Interests

- Ocean biogeochemical processes and marine ecosystems; trace-metal and nutrient cycling
- Iron biogeochemistry, redox chemistry, organic ligands, scavenging, and sedimentary fluxes
- Marine biogeochemical and ecosystem modeling; coupling circulation, chemistry, and biology
- Marine carbon cycling, benthic–pelagic coupling, and climate-change feedbacks
- Ocean observations, data synthesis, and model–data comparison

Publications

Floback, A. E., Pham, A. L.-D., Damien, P., Odendahl, C. E., Weiske, J., Thomas, K. E., Gaffney, J., John, S. G., Bianchi, D., & Moffett, J. W. (2026). Iron supply from the Oregon margin to the ocean dominated by hypoxia-dependent particles. *Proceedings of the National Academy of Sciences*, 123(18), e2532929123. <https://doi.org/10.1073/pnas.2532929123>

Pham, A. L.-D., Floback, A., Evans, N., Till, C., Till, R., Moffett, J. W., Barbeau, K., & Bianchi, D. (2026). Exploring iron processes in the California Current System through a regional dataset of dissolved iron and organic ligands measurements. *EGUsphere* [preprint], egusphere-2026-4502. <https://doi.org/10.5194/egusphere-2026-4502>

Li, D., Ruan, X., Pham, A. L.-D., Damien, P., & Bianchi, D. (2025). Role of eddies in primary production in the California Current System. *Geophysical Research Letters*, 52, e2025GL118614. <https://doi.org/10.1029/2025GL118614>

Schultz, C., Luo, J. Y., Brady, D. C., Fulweiler, R. W., Long, M. H., Petrik, C. M., Testa, J. M., Benway, H. M., Burdige, D., Cecchetto, M. M., Elegbede, I., Evans, N., Frenzel, A., Gillen, K., Herbert, L. C., Hirsh, H. K., Lessin, G., Levin, L., Maiti, K., Malkin, S., Mincks, S. L., Nmor, S., Pham, A., Pinckney, J., Rabouille, C., Rahman, S., Rakshit, S., Ray, N. E., Sasaki, D. K., Siedlecki, S. A., Somes, C., Stubbins, A., Sulpis, O., Trevisan, C., Xu, Y., & Yin, H. (2025). Elucidating the role of marine benthic carbon in a changing world. *Global Biogeochemical Cycles*, 39, e2025GB008643. <https://doi.org/10.1029/2025GB008643>

Siedlecki, S., Nmor, S., Lessin, G., Kearney, K., Rakshit, S., Petrik, C., Luo, J., Schultz, C., Sasaki, D., Gillen, K., Pham, A., Somes, C., Brady, D., Testa, J., Rabouille, C., Elegbede, I., & Sulpis, O. (2025). Sediment Biogeochemistry Model Intercomparison Project (SedBGC_MIP): motivation and guidance for its experimental design. *EGUsphere* [preprint], egusphere-2025-1846. <https://doi.org/10.5194/egusphere-2025-1846>

Buesseler, K. O., Bianchi, D., Chai, F., Cullen, J. T., Estapa, M., Hawco, N., John, S., McGillicuddy, D. J., Morris, P. J., Nawaz, S., Nishioka, J., Pham, A., Ramakrishna, K., Siegel, D. A., Smith, S. R., Steinberg, D., Turk-Kubo, K. A., Twining, B. S., Webb, R. M., Wells, M., White, A., Xiu, P., & Yoon, J.-E. (2024). Next steps for assessing ocean iron fertilization for marine carbon dioxide removal. *Frontiers in Climate*, 6, 1430957. <https://doi.org/10.3389/fclim.2024.1430957>

Pham, A. L.-D., Damien, P., McCoy, D., Mar, M., Kessouri, F., McWilliams, J. C., et al. (2024). The shelf-to-basin transport of iron from the northern U.S. West Coast to the Pacific Ocean. *Global Biogeochemical Cycles*, 38, e2023GB008029. <https://doi.org/10.1029/2023GB008029>

Robinson, D., Pham, A. L. D., Yousavich, D. J., Janssen, F., Wenzhöfer, F., Arrington, E. C., Gosselin, K. M., Sandoval-Belmar, M., Mar, M., Valentine, D. L., Bianchi, D., & Treude, T. (2024). Iron “ore” nothing: benthic iron fluxes from the oxygen-deficient Santa Barbara Basin enhance phytoplankton productivity in surface waters. *Biogeosciences*, 21, 773–788. <https://doi.org/10.5194/bg-21-773-2024>

Pham, A. L.-D., Aumont, O., Ratnarajah, L., & Tagliabue, A. (2022). Examining the interaction between free-living bacteria and iron in the global ocean. *Global Biogeochemical Cycles*, 36, e2021GB007194. <https://doi.org/10.1029/2021GB007194>

Pham, A. L.-D., & Ito, T. (2021). Anthropogenic iron deposition alters the ecosystem and carbon balance of the Indian Ocean over a centennial timescale. *Journal of Geophysical Research: Oceans*. <https://doi.org/10.1029/2020JC016475>

Pham, A. L.-D., & Ito, T. (2019). Ligand binding strength explains the distribution of iron in the North Atlantic Ocean. *Geophysical Research Letters*, 46. <https://doi.org/10.1029/2019GL083319>

Pham, A. L.-D., & Ito, T. (2018). Formation and maintenance of the GEOTRACES subsurface dissolved iron maxima in an ocean biogeochemistry model. *Global Biogeochemical Cycles*, 32. <https://doi.org/10.1029/2017GB005852>

Manuscripts in Preparation

Pham, A. L.-D., & Aumont, O. Zooplankton demand modulates iron recycling and trophic interactions. Manuscript in preparation for *Biogeosciences*.

Research Funding & Sponsored Projects

2024–2027 NSF OCE-2343357, “Iron Protection, Removal, and Transport along Anoxic Margins and Impacts on the Open Ocean.” Principal Investigator.

Pending NSF Collaborative Research, “Factors controlling iron distribution on the Southern Oregon shelf.” Co-Principal Investigator with Daniele Bianchi, James Moffett, and Kristen Buck.

Academic & Research Experience

10/2021–present Postdoctoral Scholar, Department of Atmospheric and Oceanic Sciences, University of California, Los Angeles, CA

01/2020–09/2021 Postdoctoral Associate, Laboratoire d’Océanographie et du Climat: Expérimentations et Approches Numériques (LOCEAN), Paris, France

08/2014–12/2019 Research Assistant, Ocean Biogeochemistry Group, Georgia Institute of Technology, Atlanta, GA

2013 Undergraduate Researcher, Hanoi University of Science; developed a 1-D ecosystem model to investigate tuna–zooplankton interactions along the Vietnam Central Coast

2012–2013 Undergraduate Research Intern, Research Institute for Management of Seas and Islands, Hanoi, Vietnam; projects on fisheries, marine environmental protection, and Ocean Health Index applications

Teaching Experience

Physics and Chemistry of the Oceans — Lecturer, proctoring, and grading for ~20 advanced undergraduate and graduate students. Topics included fluid dynamics, thermodynamics, aquatic chemistry, gas transfer, photosynthesis and respiration, ocean circulation, and biogeochemical cycling.

Introduction to Environmental Sciences — Lecturer, proctoring, and grading for ~50 students. Topics included Earth-system components, interactions among atmosphere, ocean, solid Earth, and biosphere, and human impacts on the environment.

Teaching & Educational Materials

Introduction to Ocean Biogeochemical Modeling — short instructional video introducing model thinking and philosophy with a focus on ocean biogeochemical dynamics. <https://www.youtube.com/watch?v=qOw7xrfKill>

Student Mentoring & Supervision

03/2025–present Cassidy Allen, Undergraduate Researcher, UCLA — Exploring the coastal ocean iron cycle through a global data compilation

03/2022–03/2024 Matthew Mar, Undergraduate Researcher, UCLA — Iron cycle along the California Current

11/2021–12/2024 De’Marcus Robinson, Ph.D. Student, UCLA — Ocean biogeochemical cycles in oxygen minimum zones in the Santa Barbara Basin

12/2020–01/2023 Philippe Cherabier, Ph.D. Student, Institut de Biologie de l’École Normale Supérieure — Integrating microbial evolutionary processes in general circulation models

06/2018–08/2018 Ashanti Holt, Undergraduate Researcher, Georgia Institute of Technology — Dissolved oxygen trends in U.S. coastal waters from 1960 to 2010

Selected Conference Presentations & Invited Seminars

April 2026 — USC Paleo/Environmental Seminar, University of Southern California
From Shelf to Basin: Mechanisms Regulating Iron Transport in the Pacific Ocean. Invited seminar.

2025 Pham A., Floback A., Damien P., Bianchi D., Moffett J. Processes controlling the shelf-to-basin transport of particulate and dissolved iron in the Pacific Ocean. Goldschmidt Conference, oral presentation.

2024 Pham A. Fate and transport of large iron fluxes from benthic sediments along the eastern Pacific coasts to the open ocean. Deep Ocean Observing Annual Meeting, oral presentation.

2023 Pham A., Damien P., McCoy D., Kessouri F., McWilliams J. C., Moffett J., Bianchi D. The shelf-to-basin transport of iron from the northern U.S. West Coast to the Pacific Ocean. Goldschmidt Conference, poster.

2022 Pham A. Evaluating biological and physical drivers of ocean iron cycling. Institute for Marine and Antarctic Studies, University of Tasmania, invited seminar.

2022 Pham A. The microbial control on the ocean cycling of trace metals under the current and changing climate. University of Vienna, invited seminar.

2022 Pham A., Aumont O. Zooplankton demand modulates the iron recycling and trophic interactions. Ocean Sciences Meeting, oral presentation.

2021 Pham A., Aumont O., Ratnarajah L., Tagliabue A. Examining the interaction between free-living bacteria and iron in the global ocean. Goldschmidt Conference, oral presentation.

2021 Pham A. Biological processes control the ocean iron distribution. Ocean Carbon and Biogeochemistry Webinar Series.

2020 Pham A. Biological processes control the ocean iron distribution. Biogeochemistry Group, Los Alamos National Laboratory, invited seminar.

2020 Pham A., Ito T., Jones D. Exploring different iron regimes in the global ocean by an unsupervised clustering technique. AGU Annual Meeting, oral presentation.

2018 Pham A., Ito T. Parameterizing multiple ligand classes improves the simulation of dissolved iron in the subtropical North Atlantic. OCB Summer Workshop, poster.

2018 Pham A., Ito T. What controls the subsurface extrema of dissolved iron in the upper ocean thermocline? Ocean Sciences Meeting, oral presentation.

Professional Service

2023–present Member, GEOTRACES Early Career Scientist Committee

02/2024 Session Convener, Ocean Sciences Meeting, “The influence of boundary currents on exchange processes between continental margins and the open ocean and biogeochemical consequences.”

07/2023 Session Convener, Goldschmidt Conference, “Marine trace element cycling from the estuaries to the open ocean (GEOTRACES).”

2021–present Moderator Crew, EarthArXiv preprint repository

2019–present Reviewer for scientific journals including Geological Magazine, Environmental Research Letters, One Earth, Biogeosciences, Biogeochemistry, Geoscientific Model Development, Environmental Science & Technology, Journal of Geophysical Research: Oceans, Global Biogeochemical Cycles, and Scientific Reports

References

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Jennifer Glass, Associate Professor, School of Earth and Atmospheric Sciences, Georgia Institute of Technology • jennifer.glass@eas.gatech.edu

Daniele Bianchi, Associate Professor, Department of Atmospheric and Oceanic Sciences, University of California, Los Angeles • dbianchi@atmos.ucla.edu